CASE STUDY

Groww Mutual Fund RAG Assistant

How I built a compliance-first AI assistant that investors and regulators can actually trust

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THE PROBLEM

Why a generic AI assistant doesn’t work for mutual fund investors

Every day, retail investors on Groww ask simple factual questions about their mutual funds — What’s the NAV? What’s the exit load? What’s the lock-in period? but the answers live scattered across dense scheme documents and factsheets that most users never read. A specific AI chatbot layered on top of this data feels like the obvious fix. It isn’t it introduces three specific risks that are simply unacceptable in a regulated financial product:

1

Hallucination risk

Blending data across similar-sounding funds produces confidently wrong answers.

2

Regulatory risk

SEBI/AMFI rules prohibit advisory-style responses even accidental ones.

3

Privacy risk

Users may paste PAN, Aadhaar, or phone numbers into a chat without thinking twice.

WHY IT MATTERED

“A wrong answer here isn’t a UX bug — it’s a compliance incident.”

For a financial assistant, trust isn’t a feature — it is the entire product.

MY APPROACH

Treat it as a trust problem, not a chatbot problem

Rather than optimizing for a clever prompt, I framed this as an infrastructure problem: what has to be true of the system structurally, not just behaviorally for a compliance team to trust it in production? That question drove three non-negotiable principles:

1

Ground everything

No answer ships without a traceable source citation — the model explains, it never invents.

2

Filter before generating

Compliance and PII checks run before a query reaches the LLM, not as a post-hoc content filter.

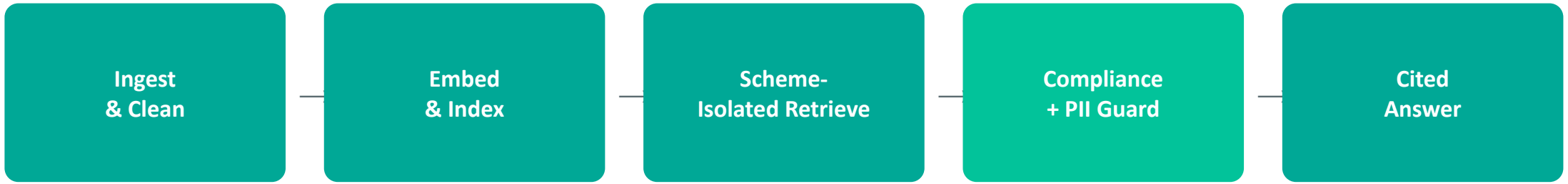
3

Make it self-healing

The system detects and repairs a stale vector index on its own, instead of depending on someone remembering to reseed it.

WHAT I BUILT

A grounded, guarded pipeline from question to cited answer



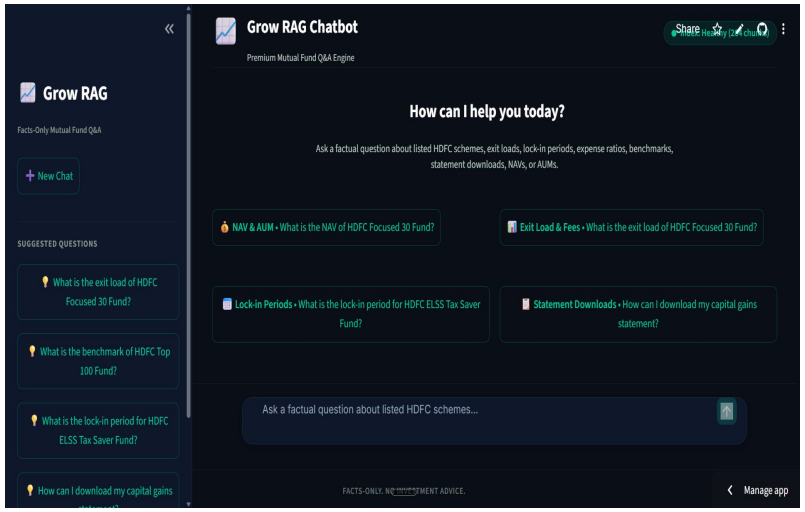
BUILT WITH

Python | FastAPI | ChromaDB | Groq (Llama 3.1) | Streamlit | bge-small-en-v1.5

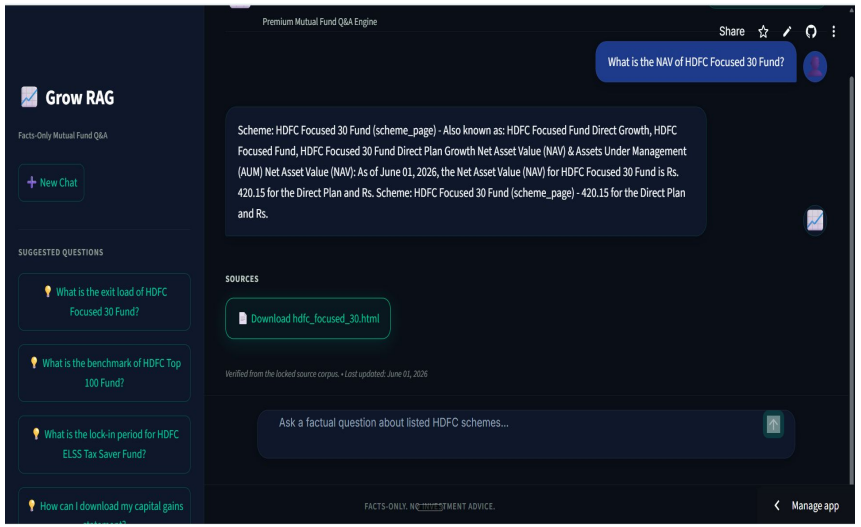
Every chunk carries its scheme + source metadata, so retrieval can never cross-contaminate between funds.

PRODUCT IN ACTION

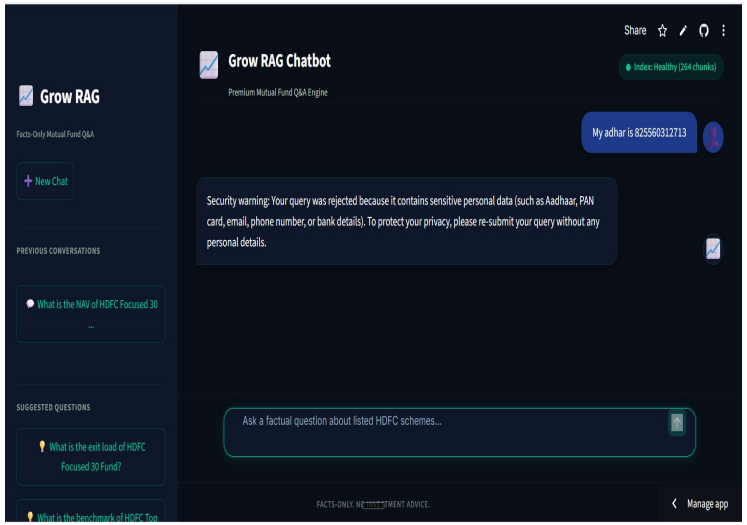
What it actually looks like to use it



Home screen — ask any factual fund question



A cited NAV answer with a downloadable source



A blocked PII query — the security guardrail in action

FROM AI CAPABILITY TO USER VALUE

Why each guardrail exists — in plain language

Technical capability	What it means for the user
Scheme-isolated retrieval	Never mixes up two similar-sounding funds
Compliance guardrails (SEBI/AMFI)	Can’t accidentally get investment advice from a chatbot
PII detection & blocking	Your personal data never reaches the AI model
Self-healing vector database	Answers stay accurate even if the backend data goes stale

KEY DECISIONS & TRADE-OFFS

Every guardrail traded some flexibility for trust

Decision	Trade-off
Strict scheme isolation over flexible fuzzy search	Eliminates cross-fund hallucination, but a query must name a specific fund
Rule-based compliance filter over LLM-only judgment	Fully auditable and predictable, but needs manual updates as new edge cases appear
Self-healing DB guard over manual reseeding	Removes on-call burden, at the cost of a small startup-time check

IMPACT

Results validated against an automated QA audit

These results come from a 9-question audit covering both factual accuracy and compliance boundaries the two things a financial AI product cannot get wrong.

264	9/9	100%	0
chunks indexed in production	**automated QA audit cases passed**	**answers with a verifiable citation**	**advisory or compliance violations**

In short: the system never once crossed into advisory territory or blended data between funds the two failures that would have made it unusable in production.

WHAT'S NEXT

If I kept building

→ Expand fund coverage beyond the current HDFC scheme set

→ Replace the rule-based compliance filter with a lightweight classifier for borderline queries

→ Add multi-turn memory so follow-ups about the same fund don’t need re-stated context

Thank you

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